

E-Commerce Revenue and Profitability Analysis

1. Project Overview

This project focuses on analysing e-commerce sales data to evaluate sales performance, revenue, and profitability. The dataset contains information about orders, products, categories, pricing, quantity, revenue, profit, customers, and locations. The data is cleaned and transformed using Excel before analysis. Power BI is used to create an interactive dashboard with KPIs, charts, tables, and slicers. The analysis highlights monthly sales trends, category performance, top products, profit margins, and customer patterns.

Objective

The objective of this project is to analyse e-commerce sales data to understand revenue and profitability performance, identify high- and low-performing products and categories, evaluate the impact of discounts, and understand customer purchasing patterns based on age and gender. The analysis also aims to compare different order channels and order statuses and identify sales trends over time. The insights generated will support data-driven decision-making and help improve sales performance and profitability.

2. Data Sources

- Source Description and Timeline: Hugging Face Datasets/E-commerce sales (January 2026 – August 2026)
- Domain: E-Commerce Sales Analytics

3. Problem Statement

- To analyse overall sales and revenue performance by examining Revenue and Profit across different products, categories, and order dates to identify key business trends.
- To evaluate profitability using Profit and Profit Margin to identify high-performing and low-performing products and categories.
- To assess the impact of discounts on Revenue, Profit, and Profit Margin to understand whether discounting strategies are supporting or reducing profitability.
- To analyse customer demographics based on Customer Age and Gender to identify customer segments that contribute most to sales and profitability.
- To compare order channels and order status to understand which sales channels perform better and how completed, cancelled, or other order statuses affect business performance.
- To identify sales and profitability trends over time using Order Date to support data-driven decisions related to product performance, pricing, discount strategies and sales channels.

4.Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Order ID	Integer	Unique identifier for each customer
Order Date	Date	Date on which the order was placed
Product ID	Integer	Unique identification code for each product
Product Name	Text	Name of the product purchased
Category	Text	Category or type of product
Unit Cost	Currency	Cost incurred to purchase or produce one unit
Unit Price	Currency	Selling price of one unit
Quantity	Whole number	Number of units purchased in an order
Gross sale	Currency	Total sales value before discounts and other deduction
Discount Percent	Percentage	Percentage of discount offered on the sale
Revenue	Currency	Actual revenue generated after applicable discounts
Commission Percent	Percentage	Percentage of sales paid as commission
Shipping Cost	Currency	Cost incurred for shipping the order
Total expense	Currency	Total expenses associated with the order
Profit	Currency	Revenue remaining after deducting expenses
Profit margin (%)	Percentage	Profit expressed as a percentage of revenue
City	Text	City where the customer/order is located
State	Text	State where the customer/order is located
Payment Method	Text	Method used by the customer to make payment
Sales Channel	Text	Channel through which the sale was made, such as online or offline
Order Status	Text	Current status of the order
Customer Age	Whole number	Age of the customer in years
Customer Gender	Text	Gender category recorded for the customer
Product Rating	Decimal number	Rating given to the product by the customer

5. Tools & Technologies

Microsoft Excel:

Used for data cleaning, preprocessing, validation, and creating Pivot Tables to perform preliminary analysis and identify key sales and profitability patterns.

Microsoft PowerBI:

Used to create interactive dashboards and visualizations for analysing Revenue, Profit, Profit Margin, Discounts, Customer Demographics, Order Channels, and Order Status. **DAX (Data Analysis Expressions)** was used to create calculated columns and measures for key business metrics and analysis.

6. Data Pre-Processing (Excel / Power Query)

Data Cleaning: Checked the dataset for missing values, duplicate records, and inconsistent entries.

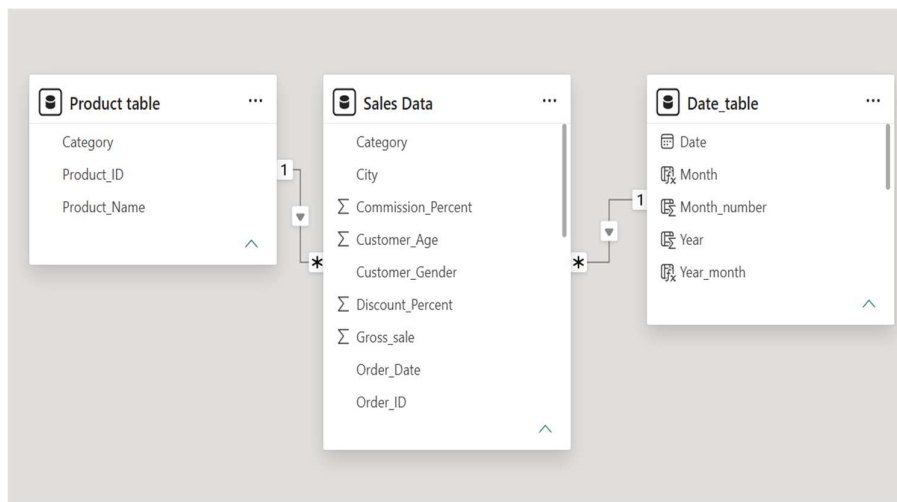
Data Formatting: Standardized column formats, particularly Order Date, numerical fields, percentages, and categorical fields.

Data Validation: Verified values in fields such as Order Status, Order Channel, Category, Customer Gender, and numerical measures.

Data Preparation: Organized the cleaned dataset and created Excel Pivot Tables for preliminary analysis and reference before importing the data into Power BI.

7. Data Modelling and DAX (Power BI)

Data Modelling: In data modelling, separate **Date** and **Product** dimension tables are created and connected them to the Sales fact table using one-to-many relationships to create a clean star schema and support efficient analysis.



Calculated Columns & DAX Measures:

Using DAX (Data Analysis Expressions) several calculated columns and measures were created to enhance the analytical insights.

- **Calculated column:**
 - Age group
 - Month
 - Month number
 - Year
- **Measures:**
 - Average Discount (%)
 - Average shipping cost
 - Number of orders
 - Profit Margin (%)

- Total profit
- Total quantity sold
- Total revenue

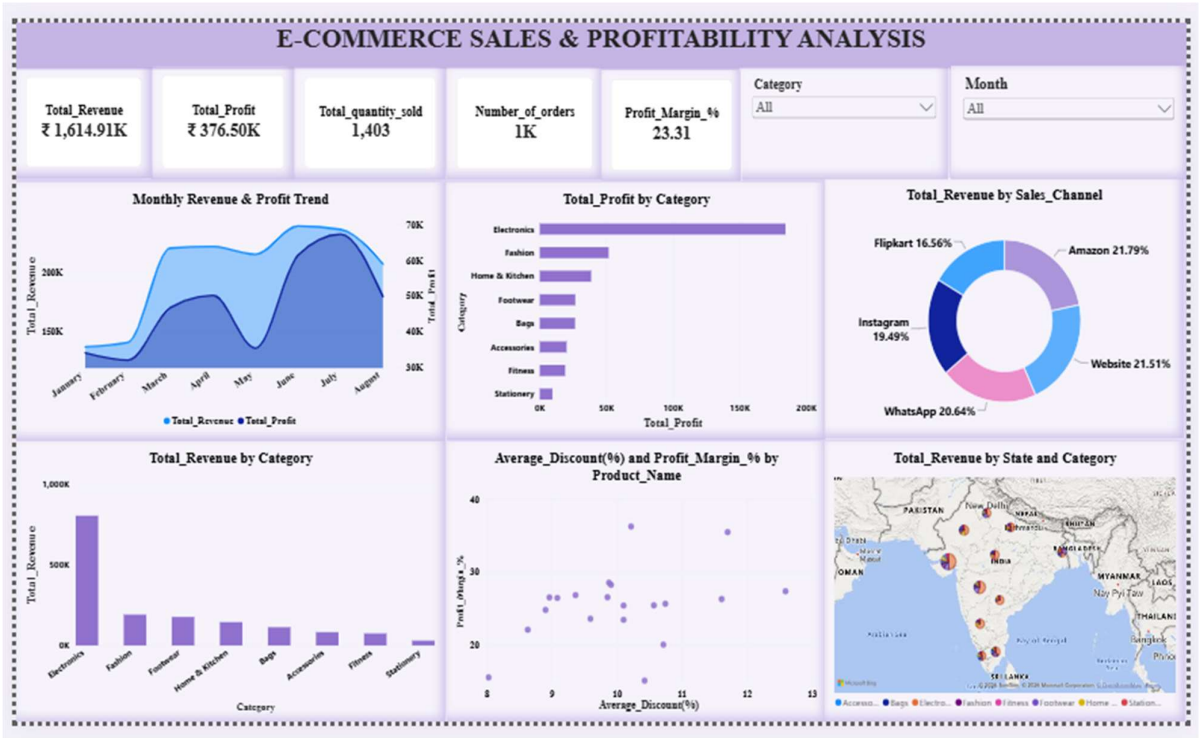
8. Analysis and Visualizations (Power BI)

Dashboard Features:

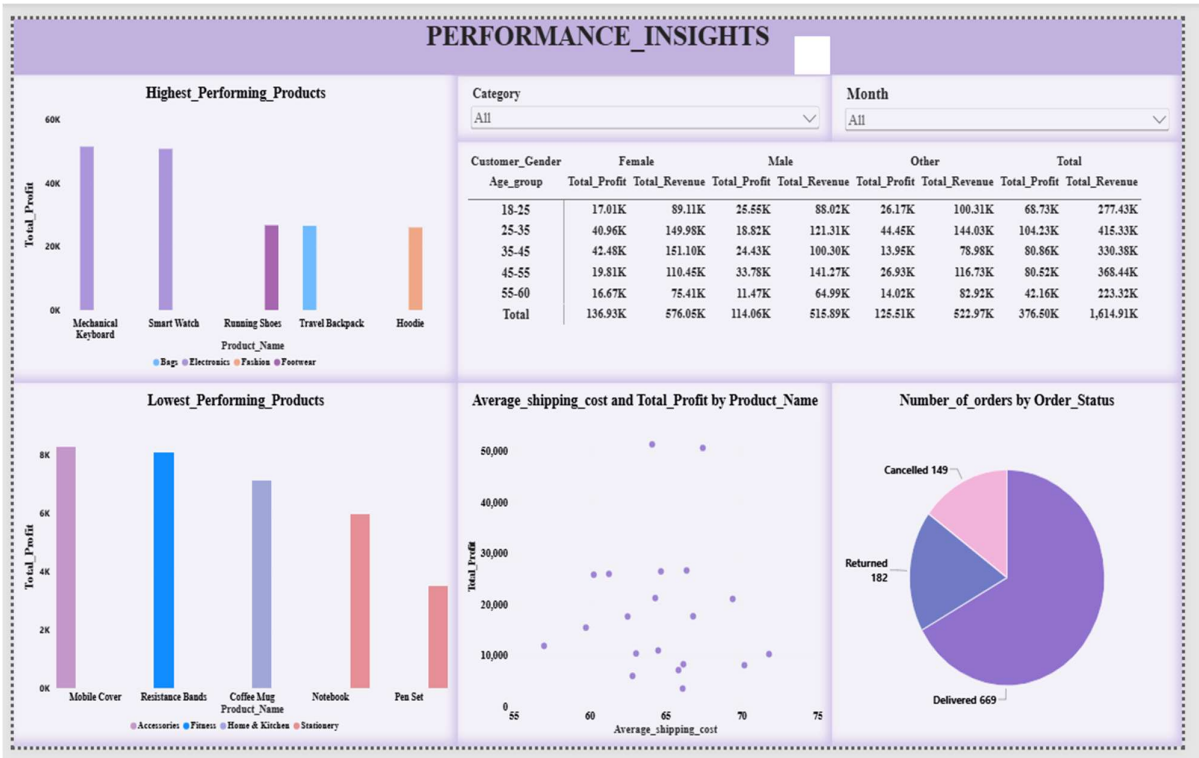
To address the identified problem statements, a series of interactive PowerBI dashboard visualizations were developed. Various charts such as line chart, bar chart, clustered column chart, donut chart, pie chart, scatter chart, matrix, KPI cards and map visualization were used to analyse sales trends, revenue, profit, profit margins, product performance, customer patterns, sales channels, and geographical performance.

1. To analyse the overall e-commerce business performance, **KPI card visuals** are used to display Total Revenue, Total Profit, Total Orders, Quantity Sold, and Profit Margin (%).
2. To analyse the monthly sales trend based on revenue and profit, an **area chart** is used to highlight changes in sales performance over time.
3. To evaluate the profitability of different product categories, a **clustered bar chart** is used to compare the profit generated by each category.
4. To analyse the revenue generated across different product categories, a **clustered column chart** is used to compare the revenue of each category. This comparison reveals the highest- and lowest-performing categories in terms of revenue and helps identify which categories contribute most to the overall sales performance.
5. To evaluate the relationship between profit margin and average discount across different products, a **scatter chart** is used with Product Name as the data point. This comparison helps identify whether higher discounts are associated with higher or lower profit margins and highlights products with relatively high or low profitability.
6. To analyse the geographical distribution of revenue across different states and product categories, a **map visual** is used. This comparison helps identify the states and categories contributing the most to overall revenue. The visual highlights regional variations in sales performance and helps identify high- and low-performing markets.
7. To analyse the contribution of different sales channels to overall revenue, a **donut chart** is used. This comparison reveals which sales channels contribute the highest and lowest share of total revenue and helps evaluate channel performance.
8. To analyse the distribution of orders across different order statuses, a **pie chart** is used. This comparison reveals the proportion of orders in each status and helps evaluate the overall order fulfilment performance.
9. To analyse revenue and profit performance across different gender and age groups, a **matrix** visual is used. This comparison reveals the most and least profitable customer segments and highlights the age and gender groups contributing the highest revenue.
10. Category and Month slicers, allowing users to quickly filter the dashboard and view specific revenue and profitability insights based on their selected category and month.

E-COMMERCE SALES & PROFITABILITY ANALYSIS



PERFORMANCE INSIGHTS



9. Insights

- The overall e-commerce business shows strong revenue performance with a healthy profit margin.
- The monthly trend shows noticeable fluctuations in both revenue and profit. Analyze the reasons behind the **June–July performance** and replicate successful strategies. Investigate the **May and August revenue declines**.
- **Prioritize Electronics**, as it is the strongest contributor to both revenue and profit. Strengthen lower-performing categories such as **Stationery and Fitness** through targeted promotions and product analysis.
- Continue investing in **Amazon and the company website**, while exploring opportunities to improve Flipkart performance.
- Discount strategies should be evaluated at the product level. Excessive discounting should be avoided where it does not generate sufficient additional sales.
- The customers **aged 25–35** generate the highest revenue and profit, making this the **strongest-performing age group**. Among the **gender** segments, **Female** customers contribute the highest overall profit.

10. Conclusion

The E-commerce Sales Analysis provides a clear understanding of the business's sales, revenue, and profitability performance. By analyzing products, categories, discounts, order channels, customer demographics, and order status, key patterns and performance differences can be identified. The analysis helps identify high-performing products and categories, profitable customer segments, effective sales channels, and the impact of discounts on profitability. The insights generated through the dashboards can support data-driven decision-making, improve sales strategies, optimize discount policies, and enhance overall business profitability.